

Overview of materials for Polycarbonate, Molded

Categories: [Polymer](#); [Thermoplastic](#); [Polycarbonate \(PC\)](#); [Polycarbonate, Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polycarbonate, Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric		English	Comments
Density	1.01 - 1.51 g/cc		0.0365 - 0.0546 lb/in ³	Average value: 1.20 g/cc Grade Count:759
	0.960 - 1.17 g/cc @Temperature 270 - 330 °C		0.0347 - 0.0423 lb/in ³ @Temperature 518 - 626 °F	Average value: 1.01 g/cc Grade Count:47
Water Absorption	0.0150 - 0.700 %		0.0150 - 0.700 %	Average value: 0.229 % Grade Count:321
Moisture Absorption at Equilibrium	0.120 - 0.350 %		0.120 - 0.350 %	Average value: 0.185 % Grade Count:214
	0.580 - 0.580 % @Temperature 100 - 100 °C		0.580 - 0.580 % @Temperature 212 - 212 °F	Average value: 0.580 % Grade Count:12
Water Absorption at Saturation	0.300 - 0.400 %		0.300 - 0.400 %	Average value: 0.309 % Grade Count:104
	0.580 - 0.580 % @Temperature 100 - 100 °C		0.580 - 0.580 % @Temperature 212 - 212 °F	Average value: 0.580 % Grade Count:1
Water Vapor Transmission	15.0 g/m ² /day		0.966 g/100 in ² /day	Average value: 15.0 g/m ² /day Grade Count:34
Oxygen Transmission	65.0 - 3150 cc-mm/m ² -24hr-atm	165 - 8000 cc-mil/100 in ² -24hr-atm		Average value: 446 cc-mm/m ² -24hr-atm Grade Count:29
Nitrogen Transmission	12.0 - 630 cc-mm/m ² -24hr-atm	30.5 - 1600 cc-mil/100 in ² -24hr-atm		Average value: 85.5 cc-mm/m ² -24hr-atm Grade Count:29
Carbon Dioxide Transmission	380 - 18900 cc-mm/m ² -24hr-atm	965 - 48000 cc-mil/100 in ² -24hr-atm		Average value: 2710 cc-mm/m ² -24hr-atm Grade Count:29
Viscosity Number	41.0 - 68.0 cm ³ /g		0.410 - 0.680 dl/g	Average value: 54.3 cm ³ /g Grade Count:47
Maximum Moisture Content	0.0200 - 0.0300		0.0200 - 0.0300	Average value: 0.0202 Grade Count:55
Linear Mold Shrinkage	0.00100 - 0.0600 cm/cm		0.00100 - 0.0600 in/in	Average value: 0.00623 cm/cm Grade Count:625
	0.00700 - 0.00700 cm/cm @Temperature 80.0 - 80.0 °C		0.00700 - 0.00700 in/in @Temperature 176 - 176 °F	Average value: 0.00700 cm/cm Grade Count:2
Linear Mold Shrinkage, Transverse	0.00260 - 0.00900 cm/cm		0.00260 - 0.00900 in/in	Average value: 0.00635 cm/cm Grade Count:193
	0.00700 - 0.00700 cm/cm @Temperature 80.0 - 80.0 °C		0.00700 - 0.00700 in/in @Temperature 176 - 176 °F	Average value: 0.00700 cm/cm Grade Count:2

Melt Flow	1.00 - 139 g/10 min	1.00 - 139 g/10 min	Average value: 17.0 g/10 min Grade Count:679
Spiral Flow	12.0 - 30.0 cm	4.72 - 11.8 in	Average value: 21.8 cm Grade Count:6

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell M	50.0 - 123	50.0 - 123	Average value: 70.6 Grade Count:134
Hardness, Rockwell R	72.0 - 124	72.0 - 124	Average value: 119 Grade Count:168
Ball Indentation Hardness	103 - 130 MPa	15000 - 18900 psi	Average value: 116 MPa Grade Count:71
Tensile Strength, Ultimate	30.0 - 105 MPa	4350 - 15200 psi	Average value: 64.2 MPa Grade Count:350
	23.0 - 51.0 MPa @Temperature 60.0 - 120 °C	3340 - 7400 psi @Temperature 140 - 248 °F	Average value: 38.1 MPa Grade Count:4
Tensile Strength, Yield	40.0 - 154 MPa	5800 - 22300 psi	Average value: 62.5 MPa Grade Count:623
Elongation at Break	3.00 - 233 %	3.00 - 233 %	Average value: 85.9 % Grade Count:596
	70.0 - 70.0 % @Temperature 60.0 - 120 °C	70.0 - 70.0 % @Temperature 140 - 248 °F	Average value: 90.0 % Grade Count:1
Elongation at Yield	4.00 - 118 %	4.00 - 118 %	Average value: 7.97 % Grade Count:381
	2.50 - 4.00 % @Temperature 60.0 - 120 °C	2.50 - 4.00 % @Temperature 140 - 248 °F	Average value: 3.17 % Grade Count:1
Modulus of Elasticity	1.58 - 6.00 GPa	229 - 870 ksi	Average value: 2.39 GPa Grade Count:441
	1.60 - 2.10 GPa @Temperature 60.0 - 120 °C	232 - 305 ksi @Temperature 140 - 248 °F	Average value: 1.83 GPa Grade Count:1
Flexural Yield Strength	27.6 - 205 MPa	4000 - 29700 psi	Average value: 90.9 MPa Grade Count:625
Flexural Modulus	0.228 - 10.0 GPa	33.0 - 1450 ksi	Average value: 2.38 GPa Grade Count:677
	1.70 - 2.20 GPa @Temperature 60.0 - 120 °C	247 - 319 ksi @Temperature 140 - 248 °F	Average value: 1.88 GPa Grade Count:3
Flexural Strain at Break	6.80 - 7.50 %	6.80 - 7.50 %	Average value: 7.06 % Grade Count:18
Flexural Strain at Yield	6.60 - 7.30 %	6.60 - 7.30 %	Average value: 7.03 % Grade Count:46
Compressive Yield Strength	76.0 - 86.2 MPa	11000 - 12500 psi	Average value: 82.4 MPa Grade Count:8
Izod Impact, Notched	0.392 J/cm - NB	0.735 ft-lb/in - NB	Average value: 8.40 J/cm Grade Count:394
	0.100 - 7.48 J/cm @Temperature -40.0 - 0.000 °C	0.187 - 14.0 ft-lb/in @Temperature -40.0 - 32.0 °F	Average value: 4.16 J/cm Grade Count:12
	2.1356 - 7.47267 J/cm @Temperature -40.0 - -30.0 °C	4.0009 - 13.9994 ft-lb/in @Temperature -40.0 - -22.0 °F	Average value: 4.16 J/cm Grade Count:6
	2.1356 - 7.47267 J/cm @Thickness 3.17 - 3.20 mm	4.0009 - 13.9994 ft-lb/in @Thickness 0.125 - 0.126 in	Average value: 4.16 J/cm Grade Count:6
Izod Impact, Unnotched	0.600 - 5340 J/cm	1.12 - 10000 ft-lb/in	Average value: 17.5 J/cm Grade Count:74
	1.50 - 2.50 J/cm @Temperature -30.0 - 0.000 °C	2.81 - 4.68 ft-lb/in @Temperature -22.0 - 32.0 °F	Average value: 2.00 J/cm Grade Count:2

Izod Impact, Notched (ISO)	6.00 - 90.0 kJ/m ²	2.86 - 42.8 ft-lb/in ²	Average value: 64.1 kJ/m ² Grade Count:180
	10.0 - 60.0 kJ/m ² @Temperature -40.0 - -20.0 °C	4.76 - 28.6 ft-lb/in ² @Temperature -40.0 - -4.00 °F	Average value: 18.8 kJ/m ² Grade Count:23
	1.20 - 150 kJ/m ² @Temperature -30.0 - -20.0 °C	0.571 - 71.4 ft-lb/in ² @Temperature -22.0 - -4.00 °F	Average value: 18.8 kJ/m ² Grade Count:71
	1.20 - 150 kJ/m ² @Thickness 3.00 - 3.20 mm	0.571 - 71.4 ft-lb/in ² @Thickness 0.118 - 0.126 in	Average value: 18.8 kJ/m ² Grade Count:71
Izod Impact, Unnotched (ISO)	21000 kJ/m ²	9990 ft-lb/in ²	Grade Count:3
	20997.9 - 20997.9 kJ/m ² @Temperature -30.0 - -30.0 °C	9991.75 - 9991.75 ft-lb/in ² @Temperature -22.0 - -22.0 °F	Grade Count:3
Charpy Impact Unnotched	4.00 J/cm ² - NB	19.0 ft-lb/in ² - NB	Average value: 24.8 J/cm ² Grade Count:247
	2.00 J/cm ² - NB @Temperature -60.0 - -20.0 °C	9.52 ft-lb/in ² - NB @Temperature -76.0 - -4.00 °F	Average value: 23.6 J/cm ² Grade Count:136
	1.60 - 1.60 J/cm ² @Temperature -30.0 - -30.0 °C	7.61 - 7.61 ft-lb/in ² @Temperature -22.0 - -22.0 °F	Average value: 23.6 J/cm ² Grade Count:1
	1.60 - 1.60 J/cm ² @Thickness 3.00 - 3.00 mm	7.61 - 7.61 ft-lb/in ² @Thickness 0.118 - 0.118 in	Average value: 23.6 J/cm ² Grade Count:1
Charpy Impact, Notched	0.300 - 11.8 J/cm ²	1.43 - 56.2 ft-lb/in ²	Average value: 4.84 J/cm ² Grade Count:227
	0.600 - 5.04 J/cm ² @Temperature -40.0 - -20.0 °C	2.86 - 24.0 ft-lb/in ² @Temperature -40.0 - -4.00 °F	Average value: 1.50 J/cm ² Grade Count:32
	1.20 - 5.00 J/cm ² @Temperature -30.0 - -30.0 °C	5.71 - 23.8 ft-lb/in ² @Temperature -22.0 - -22.0 °F	Average value: 1.50 J/cm ² Grade Count:62
	1.20 - 5.00 J/cm ² @Thickness 3.00 - 3.00 mm	5.71 - 23.8 ft-lb/in ² @Thickness 0.118 - 0.118 in	Average value: 1.50 J/cm ² Grade Count:62
Gardner Impact	36.2 - 169 J	26.7 - 125 ft-lb	Average value: 67.1 J Grade Count:56
	>= 36.1552 J @Temperature -30.0 - -18.0 °C	>= 26.6667 ft-lb @Temperature -22.0 - -0.400 °F	Average value: 36.2 J Grade Count:5
Tensile Impact Strength	546 - 630 kJ/m ²	260 - 300 ft-lb/in ²	Average value: 562 kJ/m ² Grade Count:18
Dart Drop, Total Energy	16.0 - 94.0 J	11.8 - 69.3 ft-lb	Average value: 65.0 J Grade Count:21
	11.0 - 16.0 J @Temperature -30.0 - -30.0 °C	8.11 - 11.8 ft-lb @Temperature -22.0 - -22.0 °F	Average value: 14.3 J Grade Count:3
Falling Dart Impact	53.8 - 1300 J	39.7 - 960 ft-lb	Average value: 698 J Grade Count:8
	75.9 - 75.9 J @Temperature -40.0 - -40.0 °C	56.0 - 56.0 ft-lb @Temperature -40.0 - -40.0 °F	Average value: 75.9 J Grade Count:1
Puncture Energy	45.0 - 65.0 J	33.2 - 47.9 ft-lb	Average value: 57.3 J Grade Count:81
	40.0 - 75.0 J @Temperature -30.0 - -30.0 °C	29.5 - 55.3 ft-lb @Temperature -22.0 - -22.0 °F	Average value: 63.6 J Grade Count:76
Tensile Creep Modulus, 1 hour	2100 - 2200 MPa	305000 - 319000 psi	Average value: 2200 MPa Grade Count:45
Tensile Creep Modulus, 1000 hours	1700 - 1900 MPa	247000 - 276000 psi	Average value: 1900 MPa Grade Count:45
Taber Abrasion, mg/1000 Cycles	10.0	10.0	Average value: 10.0 Grade Count:13

Electrical Properties	Metric	English	Comments
Electrical Resistivity	10200 - 1.00e+18 ohm-cm	10200 - 1.00e+18 ohm-cm	Average value: 1.88e+16 ohm-cm Grade Count:366
Surface Resistance	100 - 4.20e+16 ohm	100 - 4.20e+16 ohm	Average value: 4.29e+15 ohm Grade Count:281
Dielectric Constant	2.70 - 3.40	2.70 - 3.40	Average value: 3.00 Grade Count:277
Dielectric Strength	11.8 - 35.0 kV/mm	300 - 889 kV/in	Average value: 28.0 kV/mm Grade Count:307
Dissipation Factor	0.000500 - 0.0150	0.000500 - 0.0150	Average value: 0.00498 Grade Count:263
Arc Resistance	0.000 - 120 sec	0.000 - 120 sec	Average value: 91.0 sec Grade Count:63
Comparative Tracking Index	85.0 - 600 V	85.0 - 600 V	Average value: 256 V Grade Count:284
Hot Wire Ignition, HWI	15.0 - 120 sec	15.0 - 120 sec	Average value: 57.7 sec Grade Count:32
High Amp Arc Ignition, HAI	30.0 - 120 arcs	30.0 - 120 arcs	Average value: 112 arcs Grade Count:32
High Voltage Arc-Tracking Rate, HVTR	10.0 - 150 mm/min	0.394 - 5.91 in/min	Average value: 55.7 mm/min Grade Count:29

Thermal Properties	Metric	English	Comments
CTE, linear	20.0 - 117 µm/m-°C	11.1 - 65.0 µin/in-°F	Average value: 66.0 µm/m-°C Grade Count:380
CTE, linear, Transverse to Flow	33.0 - 90.0 µm/m-°C	18.3 - 50.0 µin/in-°F	Average value: 66.7 µm/m-°C Grade Count:215
Specific Heat Capacity	1.25 - 2.28 J/g-°C	0.299 - 0.545 BTU/lb-°F	Average value: 1.65 J/g-°C Grade Count:58
Thermal Conductivity	0.163 - 2.00 W/m-K	1.13 - 13.9 BTU-in/hr-ft²-°F	Average value: 0.216 W/m-K Grade Count:129
Melting Point	224 - 315 °C	435 - 599 °F	Average value: 282 °C Grade Count:10
Maximum Service Temperature, Air	116 - 150 °C	240 - 302 °F	Average value: 124 °C Grade Count:97
Hot Ball Pressure Test	90.0 - 144 °C	194 - 291 °F	Average value: 131 °C Grade Count:109
Deflection Temperature at 0.46 MPa (66 psi)	57.2 - 208 °C	135 - 406 °F	Average value: 139 °C Grade Count:432
Deflection Temperature at 1.8 MPa (264 psi)	77.8 - 187 °C	172 - 369 °F	Average value: 128 °C Grade Count:692
Vicat Softening Point	100 - 218 °C	212 - 424 °F	Average value: 146 °C Grade Count:416
Minimum Service Temperature, Air	-40.0 °C	-40.0 °F	Average value: -40.0 °C Grade Count:43

Glass Transition Temp, Tg	142 - 152 °C	288 - 306 °F Average value: 146 °C Grade Count:70	
UL RTI, Electrical	75.0 - 154 °C	167 - 310 °F Average value: 120 °C Grade Count:204	
UL RTI, Mechanical with Impact	75.0 - 130 °C	167 - 266 °F Average value: 117 °C Grade Count:184	
UL RTI, Mechanical without Impact	75.0 - 150 °C	167 - 302 °F Average value: 120 °C Grade Count:184	
Flammability, UL94	HB - 5VA	HB - 5VA	Grade Count:541
Flash Point	450 - 550 °C	842 - 1020 °F Average value: 510 °C Grade Count:66	
Oxygen Index	22.0 - 40.0 %	22.0 - 40.0 % Average value: 29.0 % Grade Count:187	
Glow Wire Test	675 - 960 °C	1250 - 1760 °F Average value: 877 °C Grade Count:192	
Ignition Temperature	530 - 550 °C	986 - 1020 °F Average value: 543 °C Grade Count:8	
Optical Properties	Metric	English	Comments
Refractive Index	1.51 - 1.59	1.51 - 1.59	Average value: 1.58 Grade Count:120
Haze	0.100 - 100 %	0.100 - 100 %	Average value: 1.79 % Grade Count:121
Gloss	3.00 - 90.0 %	3.00 - 90.0 %	Average value: 47.3 % Grade Count:7
Transmission, Visible	20.0 - 93.0 %	20.0 - 93.0 %	Average value: 87.3 % Grade Count:230
UV Transmittance	88.0 %	88.0 %	Average value: 88.0 % Grade Count:32
Processing Properties	Metric	English	Comments
Processing Temperature	249 - 320 °C	480 - 608 °F Average value: 281 °C Grade Count:49	
Nozzle Temperature	227 - 343 °C	441 - 650 °F Average value: 292 °C Grade Count:128	
Melt Temperature	210 - 343 °C	410 - 650 °F Average value: 290 °C Grade Count:384	
Mold Temperature	32.2 - 160 °C	90.0 - 320 °F	Average value: 88.9 °C Grade Count:397
Injection Velocity	200 mm/sec	7.87 in/sec	Average value: 200 mm/sec Grade Count:115
Drying Temperature	70.0 - 130 °C	158 - 266 °F Average value: 118 °C Grade Count:266	
Moisture Content	0.0100 - 0.0400 %	0.0100 - 0.0400 %	Average value: 0.0203 % Grade Count:107
Dew Point	-28.9 - -17.8 °C	-20.0 - 0.000 °F	Average value: -25.2 °C Grade Count:15
Drying Air Flow Rate	22.7 - 28.3 l/min	0.800 - 1.00 ft³/min (CFM)	Average value: 25.5 l/min Grade Count:5
Injection Pressure	5.52 - 152 MPa	800 - 22000 psi	Average value: 71.1 MPa Grade Count:54

Vent Depth	0.00127 - 0.00762 cm	0.000500 - 0.00300 in	Average value: 0.00283 cm Grade Count:46
Cushion	0.635 cm	0.250 in	Average value: 0.635 cm Grade Count:5

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.